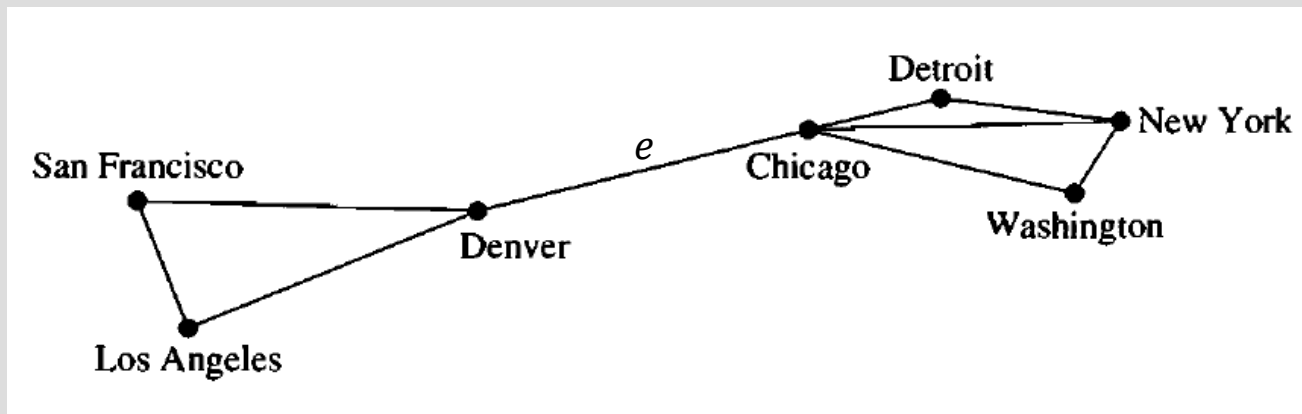


GRAPHS

CSI 219 – Discrete Mathematics

GRAPH TERMINOLOGY (UNDIRECTED GRAPH)

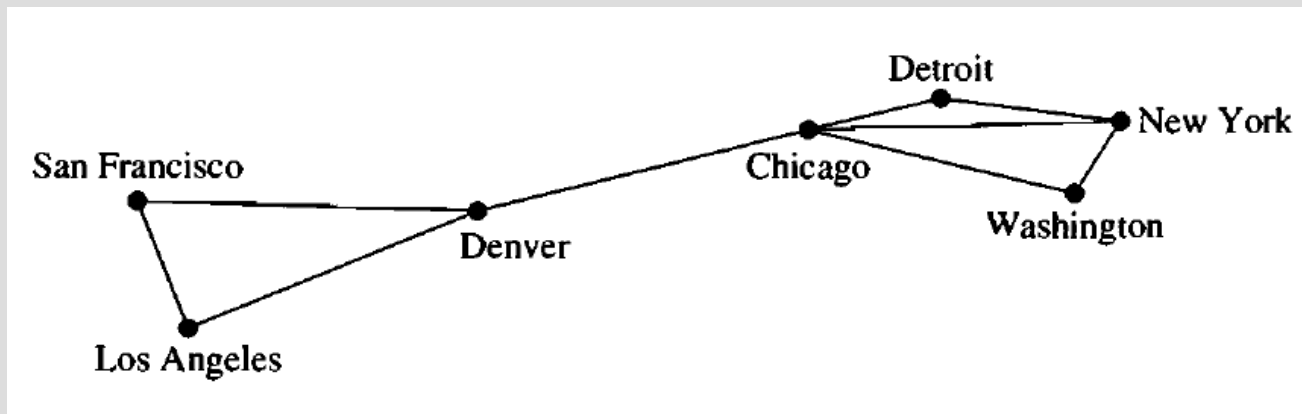


- Vertices Denver and Chicago are **adjacent** to each other
- The edge e is said to be **incident** with the vertices
- What are the adjacent pairs of vertices in this graph?

DEGREE OF A VERTEX (UNDIRECTED GRAPH)

- The degree of a vertex in an undirected graph is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex.

GRAPH TERMINOLOGY (UNDIRECTED GRAPH)



- The **degree** of the vertex Chicago is 4
 - We write $\deg(\text{Chicago}) = 4$
- $\deg(\text{Denver}) = 3$
- What are the degrees of the remaining vertices?

EXAMPLE

- What are the degrees of the vertices in the graphs G and H displayed in Figure 1?

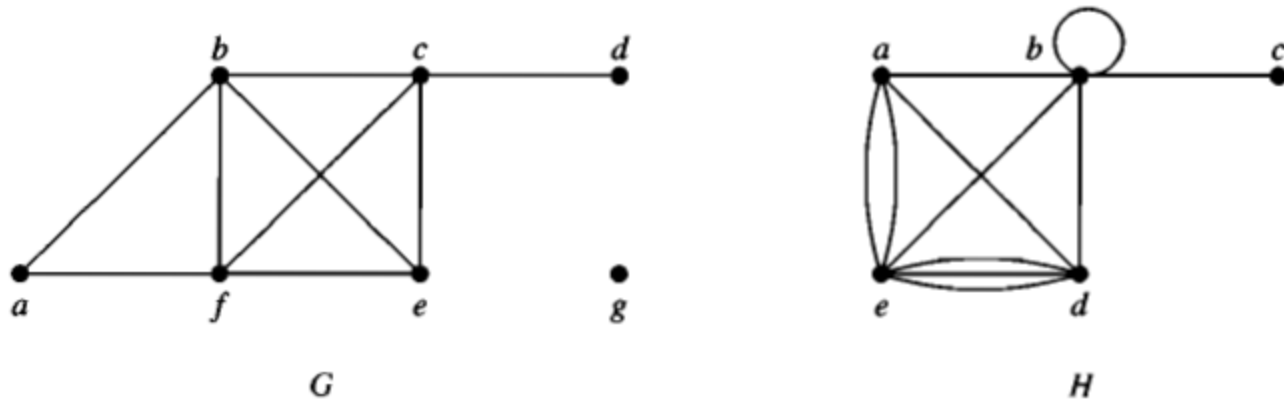


FIGURE 1 The Undirected Graphs G and H .

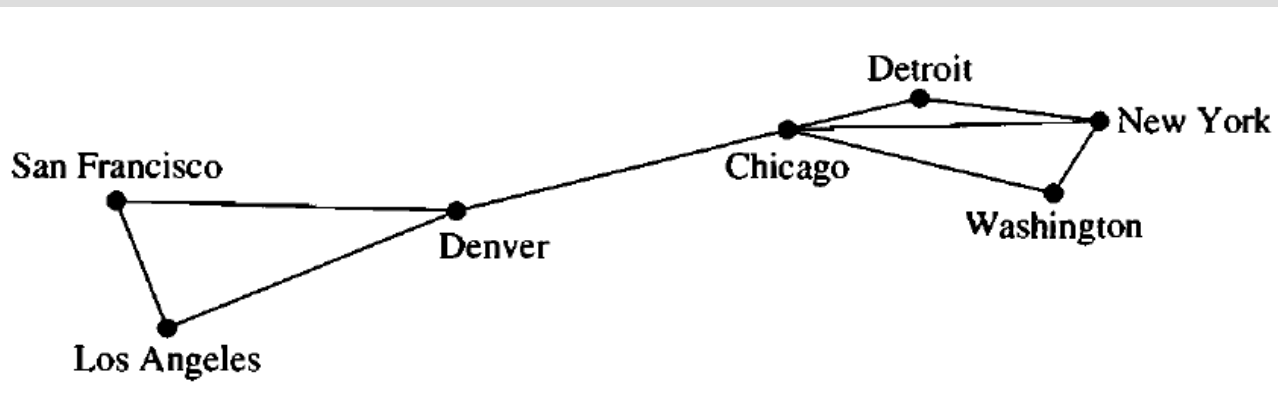
HANDSHAKING THEOREM

- Let $G = (V, E)$ be an undirected graph with e edges. Then,

$$2e = \sum_{v \in V} \deg(v)$$

- This applies to any kind of undirected graph! (Note that this applies even if multiple edges and loops are present.)

HANDSHAKING THEOREM



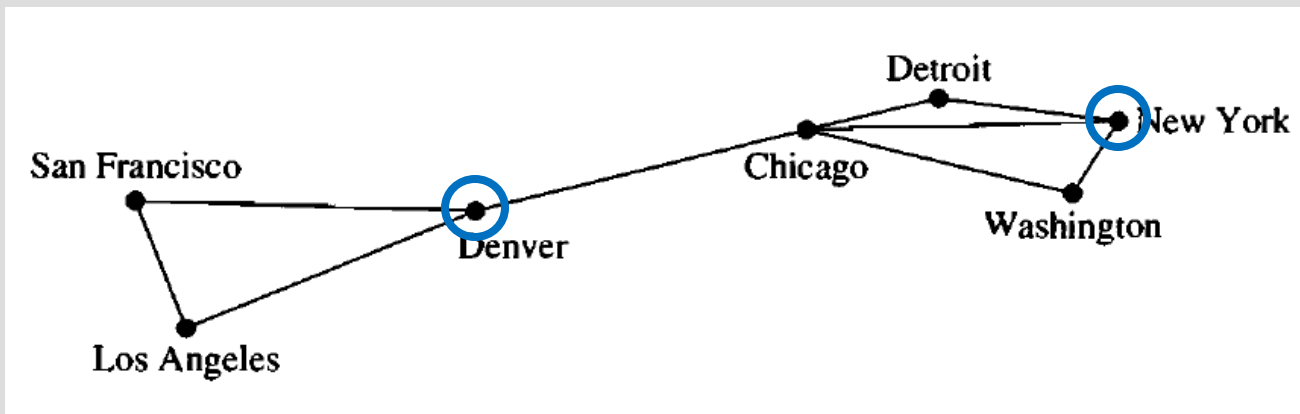
- No. of edges in this graph = 9
- Sum of degrees = $18 = 2 \times 9$
- Coincidence?

QUICK EXERCISE

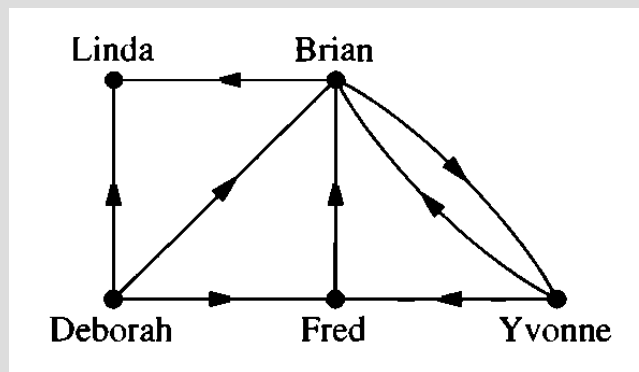
How many edges are there in a graph with 10 vertices each of degree six?

ANOTHER THEOREM

- An undirected graph has an even number of vertices of odd degree.
- Really??



GRAPH TERMINOLOGY (DIRECTED GRAPH)

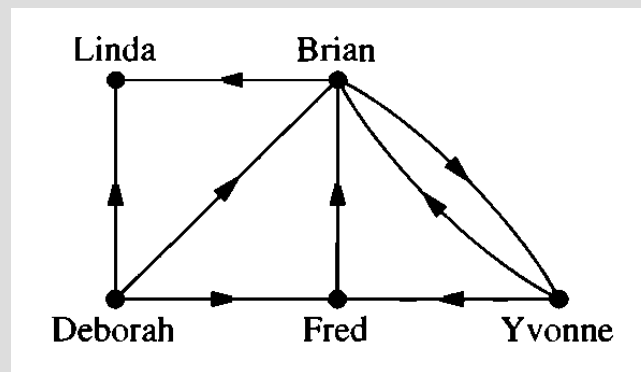


- Vertices Deborah is **adjacent** to Brian, and Brian is **adjacent** from Deborah
- Deborah is the **initial** vertex of the edge, and Brian is the **terminal** vertex of the edge

DEGREE OF A VERTEX (DIRECTED GRAPH)

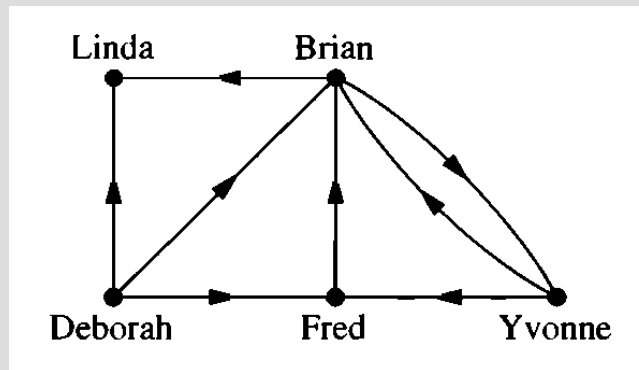
- In a graph with directed edges the *in-degree of a vertex* v , denoted by $\deg^-(v)$, is the number of edges with v as their terminal vertex.
- The *out-degree of* v , denoted by $\deg^+(v)$, is the number of edges with v as their initial vertex.
- Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of this vertex.

GRAPH TERMINOLOGY (DIRECTED GRAPH)



- The **in-degree** of Brian is 3
 - $\deg^-(Brian) = 3$
- The **out-degree** of Brian is 2
 - $\deg^+(Brian) = 2$

HANDSHAKING THEOREM (DIRECTED GRAPH)



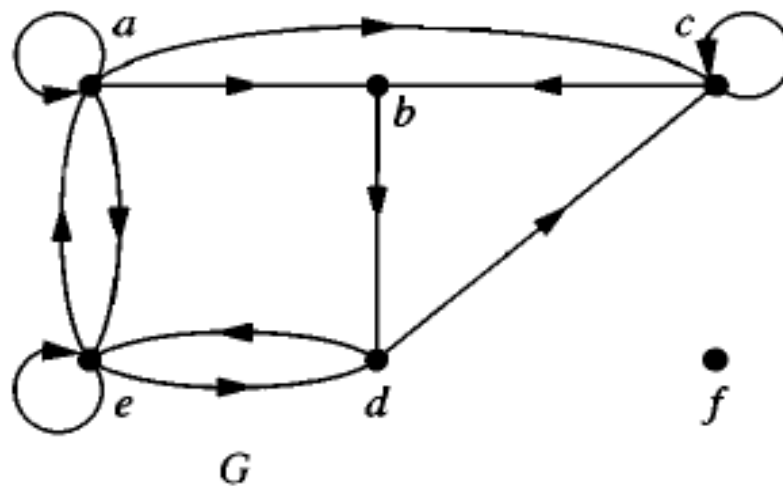
- The sum of in-degrees is 8
- The sum of out-degrees is 8
- No. of edges is 8
- Coincidence?

HANDSHAKING THEOREM (DIRECTED GRAPH)

- Let $G = (V, E)$ be a graph with directed edges. Then

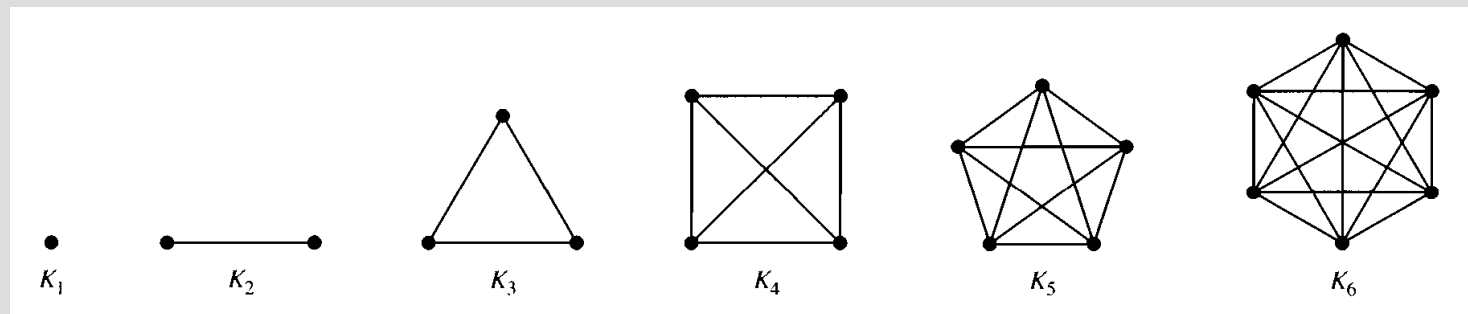
$$\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|$$

EXAMPLE



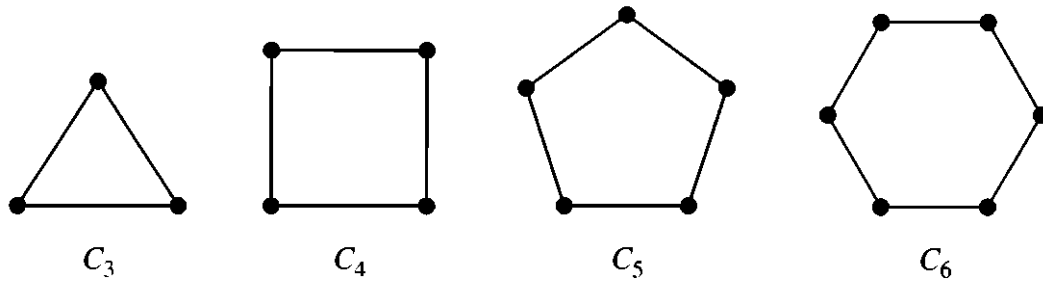
SOME SPECIAL SIMPLE GRAPHS

COMPLETE GRAPH



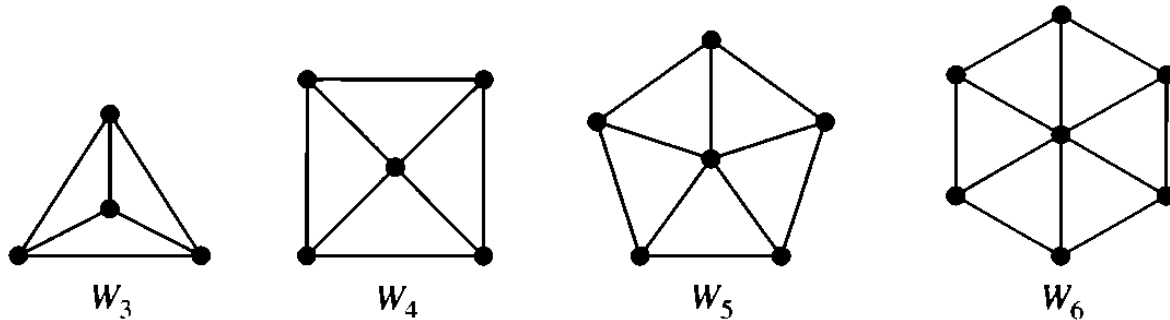
- Simple graph where there is exactly one edge between each pair of **distinct** vertices.

CYCLES



- Denoted by C_n
- n edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}, \{v_n, v_1\}$
- What is the minimum value of n ?

WHEELS



- An additional vertex to cycle, and connect every vertex of the cycle to the new one by an edge
- What is the number of vertices in W_n ?

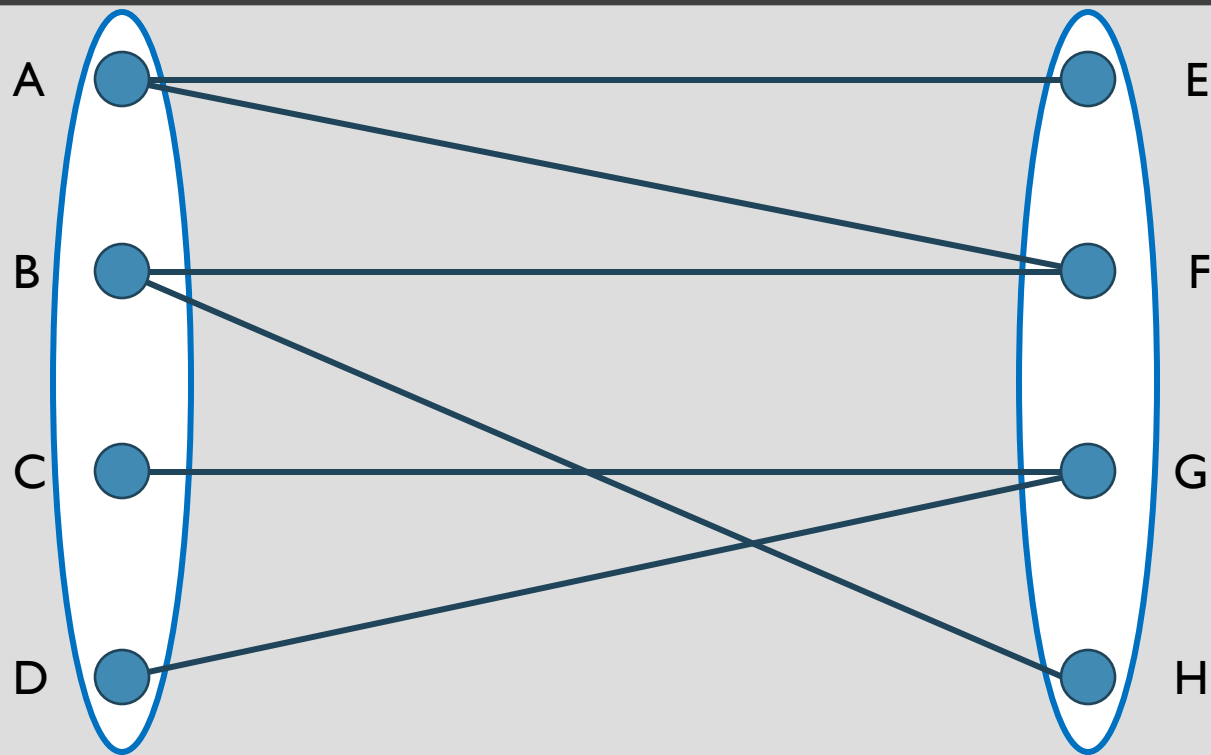
PRACTICE PROBLEM

- Chapter-9.2(Rosen,6th edition)
- Exercise- 3,4,5,10,20(a,d,e)

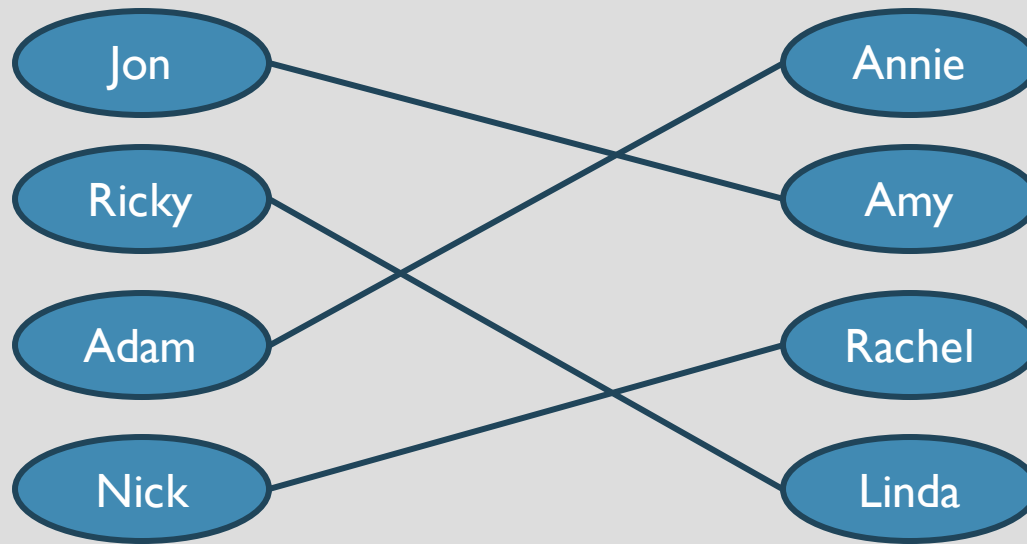
BIPARTITE GRAPH

- A simple graph G is called *bipartite* if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 (so that no edge in G connects either two vertices in V_1 or two vertices in V_2). When this condition holds, we call the pair (V_1, V_2) a *bipartition* of the vertex set V of G .

BIPARTITE GRAPH

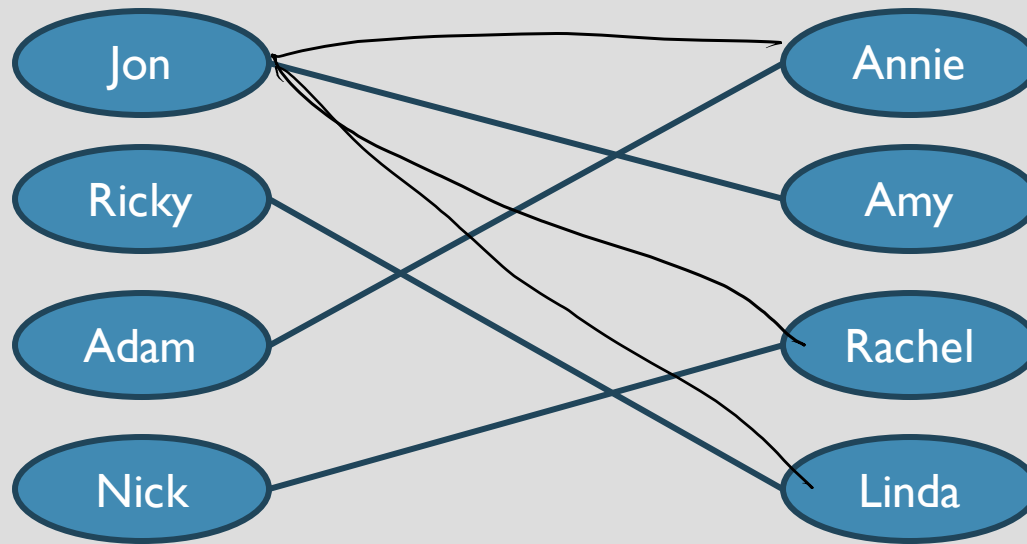


BIPARTITE GRAPH



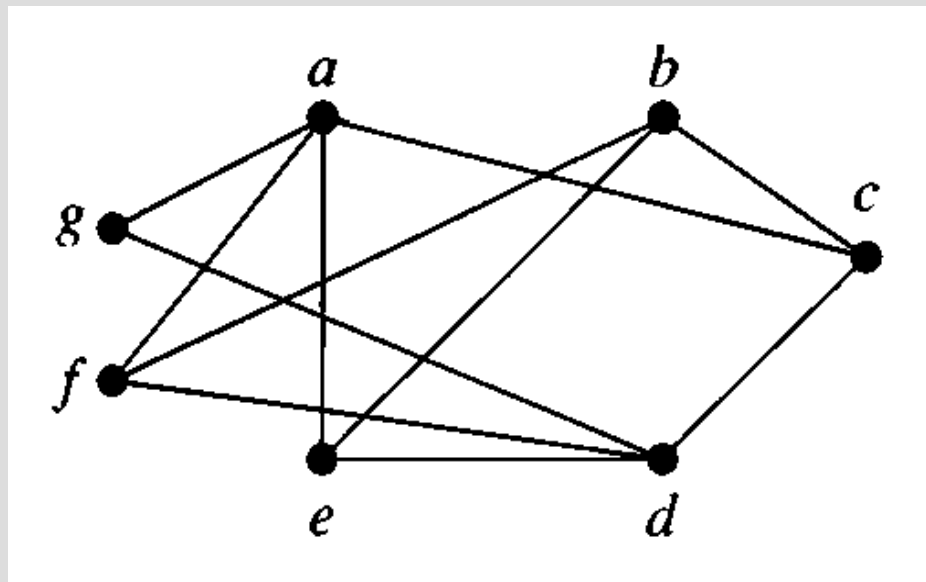
An edge represents that the two persons are married

BIPARTITE GRAPH

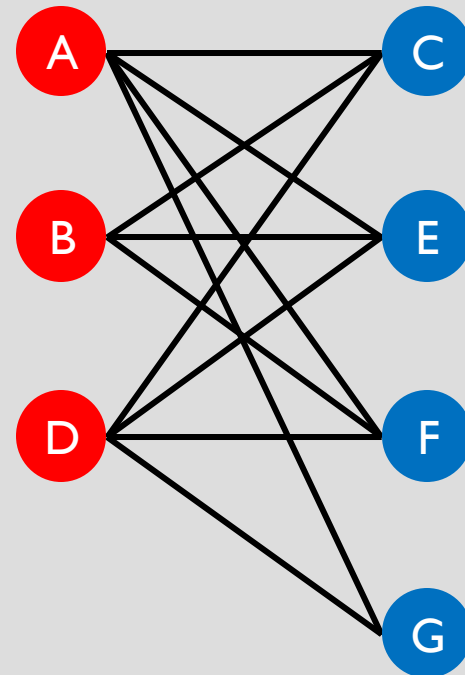
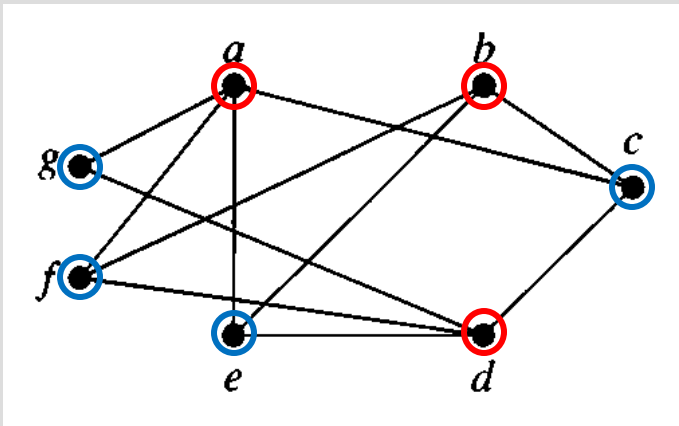


Left hand side represents actors and the right hand side represents directors
Assume that none of the actors have worked together in a movie
And none of the directors have collaborated and directed a movie

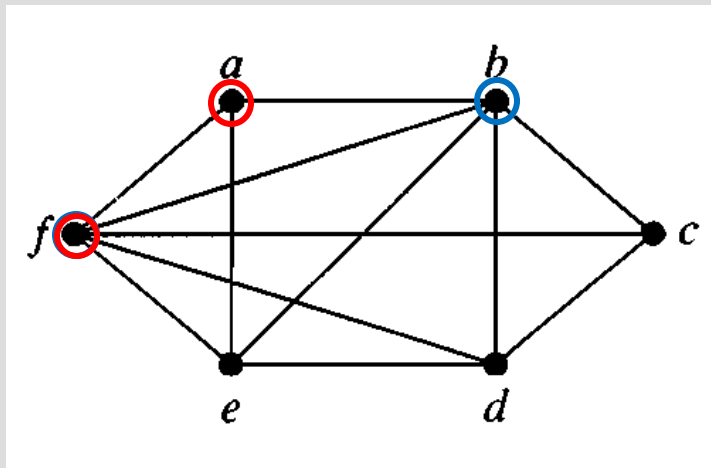
IS THIS GRAPH BIPARTITE?



IS THIS GRAPH BIPARTITE?



IS THIS GRAPH BIPARTITE?



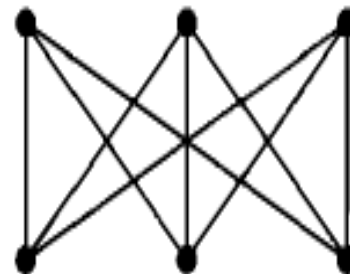
PRACTICE PROBLEM

- Exercise- 21-25 (Chapter-9.2)

COMPLETE BIPARTITE GRAPH



$K_{2,3}$



$K_{3,3}$